

JavaScript, React & MySQL

Practical Interview Question Paper

JavaScript (10 Questions)

1. Write a JavaScript program to find the **second largest element** in an array.

- Input: [25, 67, 12, 89, 45, 90]

2. Write a JavaScript program to **reverse a string** without using reverse().

- Input: "JavaScript"

3. Write a JavaScript program to check whether a given number is a **Palindrome**.

Input: 121. Output: Palindrome.

4. Write a JavaScript program to **count the number of vowels** in a given string.

- Input: "Frontend Development"

5. Write a JavaScript program to find the **duplicate elements** in an array.

- Input: [10, 20, 30, 20, 40, 10, 50]

6. Write a JavaScript program to find the **sum of all even numbers** in an array.

- Input: [10, 15, 20, 25, 30, 35]

7. Write a JavaScript program to **count the number of words** in a sentence.

- Input: "Learning JavaScript is very interesting"

8. Create a JavaScript webpage where the user enters **Principal Amount, Rate and Time** and display Simple Interest and Total Amount when the button is clicked.

9. Create a JavaScript program using an **array of objects** to store 5 student records. Each object should contain Student Name, Roll Number and Marks. Display only students who scored more than 75 marks.

10. Create a JavaScript program to find the **most repeated element** in an array.

- Input: [10, 20, 10, 30, 20, 10, 40]

React (10 Questions)

11. Create a React component to display **Student Details using Props**. Display Student Name, Roll Number, Department and CGPA.

12. Create a React **Counter Application** using useState with Increment, Decrement and Reset buttons.

13. Create a React **Student List** using an array and map(). Display Student Name, Department and Marks. Display at least 5 students.

14. Create a React **Product Card** component displaying Product Image, Product Name, Price, Rating and Add to Cart Button.
15. Create a React program using useState to update an **object** containing name, email and city. Create separate buttons to update each value.
16. Create a React program using useState to update an **array**. Display 5 names and provide a button to add a new name.
17. Create a React **Registration Form** using useState with Name, Email, Password and Course. On submit, display the entered values.
18. Create a React **Shopping Cart** using useState. Display 3 products with Product Name, Price and Add to Cart. Display number of items and total price.
19. Create a React component using useEffect to **fetch data from an API** and display ID, Name and Email in a table.
20. Create a React **Login Page** using useState with Email, Password and Login Button. Display Login Successful for correct details, otherwise Invalid Email or Password.

MySQL (10 Questions)

21. Given Student(student_id, student_name, marks, course_id) and Course(course_id, course_name), display students who scored **more than 75 marks**.
22. Display the student who has the **highest marks** using a subquery.
23. Display the **total number of students in each course** using GROUP BY.
24. Display courses having **more than 3 students** using GROUP BY and HAVING.
25. Display Student Name, Course Name and Marks using **INNER JOIN**.
26. Display **all courses and their students** using LEFT JOIN, including courses with no students.
27. Display students whose marks are **greater than the average marks** using a subquery.
28. Display Highest Marks, Lowest Marks, Average Marks and Total Students using aggregate functions.
29. Create a query to display the **Palindrome numbers** from a number column. Example data: 121, 123, 1331, 456. Expected output: 121, 1331.
30. Create a **Student Performance Report** displaying Student ID, Student Name, Course Name, Marks and Result. If marks are ≥ 40 display Pass, otherwise Fail. Use CASE and JOIN.